NOTRE DAME UNIVERSITY BANGLADESH

Machine Learning Assignment-01

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Course Title: Machine Learning

Assignment Topic: Reporting about selected research paper

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1 Paper Information

Title: Beyond Performance: Explaining and Ensuring Fairness in Student Academic Performance Prediction with Machine Learning
Authors: Kadir Kesgin, Salih Kiraz, Selahattin Kosunalp and Bozhana Stoycheva
Year: 2025
Source: MDPI Access

2 Problem Description

The possibility of bias in educational data mining is the main issue this research attempts to solve. In order to give early interventions, schools frequently employ machine learning to forecast student progress. These algorithms might, however, unjustly punish students based on delicate characteristics like gender or socioeconomic background if they are based on skewed data. This is crucial because unjust AI may result in kids not receiving the resources they are entitled to.
Keywords: student performance prediction, logistic regression, explainable AI (XAI), SHAP, adversarial debiasing, SMOTE, educational data mining, machine learning in education, model interpretability

3 Machine Learning Type

Supervised Learning is used in this study. Because the researchers employ labeled datasets with known student performance outcomes, this learning style is appropriate. These established labels are mapped to input features by the trained model.

4 Model / Algorithm Used

The researchers tested a number of algorithms, such as **Random Forest** and **XGBoost**.

- **Inputs:** Information like family status, prior grades, and student demographics.
- **Outputs:** Estimated academic achievement (e.g., grade level or pass/fail).
- **Functionality:** These models generate a set of rules for making decisions. To provide a final, more precise prediction, a Random Forest, for instance, aggregates the ”opinions” of numerous tiny decision trees.

5 Evaluation Metrics

To gauge how frequently the model is accurate, the study employs common metrics such as **Accuracy** and the **F1-Score**. Importantly, fairness using the demographic information are also used. These are employed to guarantee that the error rate of the model is constant across various demographic groupings.
In Brief, they have used Random Forest, XGBoost & Linear Regression.
Random Forest is described as a bagging ensemble method. It can be compared to a "committee" of specialists coming to a consensus. During training, it deliberately builds several separate decision trees. A random subset of the data is used to construct each tree. It aggregates (combines) the forecasts of all the trees rather than depending on just one. Typically, a majority vote from every tree is required for classification.
XGBoost (XGB) represents extreme Gradient Boosting. XGBoost creates trees sequentially, whereas Random Forest generates trees simultaneously. It is a gradient boosting framework in which every new tree is specially created to fix the mistakes produced by the ones that came before it.

6 Overfitting, Underfitting, Bias & Variance

In the research paper **Bias** is heavily emphasized in the study. It mentions that "algorithmic bias" occurs when models favor particular groups. To address this, the authors employ fairness-processing strategies that stop the model for making unfair choices during training.

7 Results and Findings

The findings demonstrate that some fairness-improving strategies greatly increase equality while marginally lowering overall accuracy. When adjusted for fairness, the XGBoost model offered a dependable compromise between justice and fairness.

8 Understanding and Learning

I now had a good understanding of how a model can be morally wrong yet mathematically correct. Understanding the precise mathematics underlying the Fairness-aware algorithms was the most challenging aspect. One key idea I picked here was that, in machine learning, accuracy is not the primary objective; we also need to think about how our predictions will affect society.

9 Conclusion

In conclusion, the study effectively shows that we can create AI-powered teaching resources that are both effective and fair. This strategy, in my opinion, works really well since it guarantees that technology benefits every student equally rather than escalating preexisting social prejudices.